Form 4 game. 15 Oct 2026 Problem solving: vector, spotted or geometric method

1. Warming · Flash [GEO] Complete: $ABCD$ is a parallelogram $\Leftrightarrow \overrightarrow{AB} = \ldots$

Rep. $\overrightarrow{AB} = \overrightarrow{DC}$ (or $\overrightarrow{AD} = \overrightarrow{BC}$).

The summits are named in the order: $[AC]$ *et* $[BD]$ are diagonals.

2. Heart · Method [REP] $A(0;0)$, $B(4;1)$, $C(5;4)$. Determine D so that $ABCD$ or a parallelogram.

Rep. $\overrightarrow{AB} = \overrightarrow{DC}$: $(4;1) = (5 - x_D; 4 - y_D)$, therefore $D(1;3)$.

3. Heart + · Write [DEM] $A(0;0)$, $B(3;1)$, $C(-1;3)$. Demonstrate that ABC is isocele rectangle in A.

Rep. $AB^2 = 10$, $AC^2 = 10$: $AB = AC$; $BC^2 = 16 + 4 = 20 = AB^2 + AC^2$: Reciprocity of Pythagoras

Two distinct justifications: length equality and right angle.

4. Back · Flash [NUM] Write 450,000 in scientific notation.

Rep. $4{,}5 \times 10^5$.

5. Red thread · Flash [SOL] How many faces, peaks and edges have a cube?

Rep. 6 faces, 8 peaks, 12 edges.

Rights of the plan

Form 5 Mon. 2 Nov. 2026 Director Vector of a right

1. Warming · Flash [GEO] Give a colinear vector to $\vec{u}\begin{pmatrix}2\\-4\end{pmatrix}$ with simpler whole coordinates.

Rep. $\begin{pmatrix}1\\-2\end{pmatrix}$ (It is divided by 2).

2. Heart · Method [GEO] Give a leading vector of (AB) for $A(1;2)$ and $B(4;8)$.

Rep. $\overrightarrow{AB}\begin{pmatrix}3\\6\end{pmatrix}$, or $\begin{pmatrix}1\\2\end{pmatrix}$ who is colinear to him.

3. Heart + · MCQ [GEO] A leading vector of the right passing through $A(0;1)$ and $B(2;-3)$ is:

A. $\begin{pmatrix}1\\-2\end{pmatrix}$ **B.** $\begin{pmatrix}1\\-4\end{pmatrix}$ **C.** $\begin{pmatrix}-4\\-2\end{pmatrix}$ **D.** $\begin{pmatrix}2\\4\end{pmatrix}$

Rep. A. $\begin{pmatrix}1\\-2\end{pmatrix}$

Ici $\overrightarrow{AB}\begin{pmatrix}2\\-4\end{pmatrix}$; The colinearity is tested by the determinant.

4. Back · Flash [STA] Notes: 8, 12, 12, 15, 18. Calculate mean, median and extent.

Rep. Average 13; median 12; extent 10.

5. Red thread · Property [DEM] ABC is right-angled at A and I is the midpoint of $[BC]$. What can be said of IA, IB and IC Why?

Rep. $IA = IB = IC$: The midpoint of the hypotenuse is the center of the circumscribed circle.

Form 6 Mar. 3 Nov. 2026 Cartesian Equation of a Right

1. Warming · Flash [SUB] Calculate $3x - 2y + 1$ for $x = 1$ and $y = -2$.

Rep. $3 + 4 + 1 = 8$.

2. Heart · Method [GEO] Right d a for equation 2 $x + 3y - 6 = 0$. Provide a guiding vector and a point d.

Rep. For $ax + by + c = 0$, $\begin{pmatrix}-b\\a\end{pmatrix}$ leads the right: $\begin{pmatrix}-3\\2\end{pmatrix}$. The point $(3;0)$ appropriate: $6 + 0 - 6 = 0$.

3. Heart + · True/false [GEO] Right or wrong? Justify. Right 2 $x + 3y - 6 = 0$ and $4x + 6y + 1 = 0$ are parallels.

Rep. True. Director Vectors $\begin{pmatrix}-3\\2\end{pmatrix}$ and $\begin{pmatrix}-6\\4\end{pmatrix}$ They are distinct: $(3;0)$ does not belong to the second right (12 $+ 1 \neq 0$).

4. Back · Table [LEC] Mode of transport of 30 students.

Transport	Bus	Bicycle	Walk	Car
Count	12	5	8	5

Give the share of students coming by bus and corner of the "bike" area of a circular diagram.

Rep. $\frac{12}{30} = 40\,\%$; $\frac{5}{30} \times 360° = 60°$.

5. Red thread · Flash [REP] On a graduated right, A a for abscisse -2 and B For Abscissus 3. Give AB and abscissa from the midpoint of the $[AB]$.

Rep. $AB = 3 - (-2) = 5$; midpoint: $\frac{-2+3}{2} = 0{,}5$.

Form 7 Game. 5 Nov. 2026 Reduced equation and slope of a non-vertical straight line

1. Warming · Flash [EQ] Isolate y in 2 $x + y - 5 = 0$.

Rep. $y = -2x + 5$.

2. Heart · Method [GEO] Give the slope and order at the origin of the right 3 $x - 2y + 4 = 0$.

Rep. $-2y = -3x - 4$, $y = \frac{3}{-}x + 2$: slope $3_{\overline{2}}$, ordered originally 2.

Trap: divide by -2 each term (signs).

3. Heart + · MCQ [GEO] The slope of the right (AB) with $A(1;2)$ and $B(5;0)$ is:

A. 2 **B.** $\frac{1}{2}$ **C.** -2 **D.** $-\frac{1}{2}$

Rep. D. $-\frac{1}{2}$

Pent $= \frac{y_B - y_A}{x_B - x_A} = \frac{-2}{4}$.

4. Back · Flash [CAL] Calculate $\frac{3}{4} + \frac{5}{6}$.

Rep. $\frac{9}{12} + \frac{10}{12} = \frac{19}{12}$.

5. Red thread · Problem [IVD] Two gears have 12 and 18 teeth.
Are they returning together to their starting position?

Rep. PPCM $(12\,;\,18) = 36$ teeth: 3 turns of the first, 2 turns of the second.

Form 8 Mon. 9 Nov. 2026 Draw a given line by an equation

1. Warming · Flash [IMG] For $y = -2x + 3$, calculate y where $x = 0$ and then when $x = 2$.

Rep. $y = 3$ and $y = -1$.

2. Heart · Method [REG] Give two points to draw the right 2 $x + y - 4 = 0$.

Rep. For $x = 0$: $y = 4$; for $x = 2$: $y = 0$. Items $(0;4)$ and $(2;0)$.

3. Heart + · Flash [GEO] What is the right equation? $x = 3$? Give a leading vector. Does she have a slope?

Rep. The vertical passing through $(3;0)$; $\begin{pmatrix}0\\1\end{pmatrix}$ it leads; it has no slope; Cartesian equation $x - 3 = 0$.

Don't write it down $= mx + p$.

4. Back · Flash [CMP] Compare $\frac{3}{7}$ and $\frac{4}{9}$

Rep. Cross products : $3 \times 9 = 27 < 28 = 4 \times 7$, so $\frac{3}{7} < \frac{4}{9}$.

5. Red thread · Flash [EQ] Solve in $\mathbb{R} : 5x - 7 = 2x + 8$.

Rep. $x = 5$

Trap: group the xdone side by changing the sign correctly.

Form 9 Mar. 10 Nov. 2026 Determine a right equation: two points, leading point and vector, point and slope

1. Warming · Flash [GEO] Slope of the right passing through $(1;3)$ and $(3;7)$?

Rep. $\dfrac{7-3}{3-1}=2$.

2. Heart · Method [GEO] Reduced equation of slope line 2 passing through $A(1;3)$.

Rep. $y-3=2(x-1)$either $y=2x+1$.

Control: $2\times 1+1=3$.

3. Heart + · Method [GEO] Right through $A(1;2)$ of a master vector $\vec{u}\begin{pmatrix}2\\-1\end{pmatrix}$: Cartesian equation by the determinant of $\overrightarrow{AM}$ and $\vec{u}$.

Rep. $M(x;y)\in d\Leftrightarrow \det\left(\overrightarrow{AM},\vec{u}\right)=0:(x-1)(-1)-(y-2)\times 2=0$, either $x+2y-5=0$.

Control: $1+4-5=0$; $\begin{pmatrix}-2\\1\end{pmatrix}$ is well colinear to $\begin{pmatrix}2\\-1\end{pmatrix}$.

4. Back · Flash [Sighs] Simplify $2^{5}\times 2^{-3}$.

Rep. $2^{5-3}=2^2=4$.

5. Red thread · Table → formula [REG] An affine function f verify: $f(0)=-1, f(1)=1, f(2)=3$. Provide $f(x)$.

Rep. Pent 2, ordered originally $-1:f(x)=2x-1$.

Form 10 Thu. 12 Nov. 2026 Rights and functions: link between registers

1. Warming · Flash [IMG] $f(x)=-3x+4$. Calculate $f(2)$ and the preimage of 10.

Rep. $f(2)=-2$; $-3x+4=10\Leftrightarrow x=-2$.

2. Heart · Method [REG] Determine the function affine f such that $f(1)=1$ and $f(3)=5$.

Rep. Pent $\dfrac{5-1}{3-1}=2$; $f(x)=2x+p$ with $2+p=1:f(x)=2x-1$.

3. Heart + · True/false [REG] True or false? Justify. All right of the plane is the curve of an affine function.

Rep. False. Counter-example: the vertical right $x=3$ is the curve of no function (number 3 would an infinity of images).

4. Back · Flash [PRB] We're launching a balanced dice with 6 faces. Likely to get a multiple of 3?

Rep. Issues 3 and 6: $\dfrac{2}{6}=\dfrac{1}{3}$.

5. Red thread · Flash [GEO] ABC is right-angled at A with $AB=6$ cm and $AC=8$ cm. Calculate BC.

Rep. $BC^2=36+64=100$, so $BC=10$ cm.

Form 11 Mon. 16 Nov. 2026 Alignment, parallelism, intersection of straight lines

1. Warming · Flash [SUB] The point $(3;5)$ Does it belong to the right $y=2x-1$?

Rep. $2\times 3-1=5$: Yes.

2. Heart · Method [GEO] Relative positions of $d:\ y=3x+1$ with $d':\ y=3x-2$, then with $d'':\ y=-x+1$.

Rep. d and d' : same slope, originally ordered separately: strictly parallel; d and d'' : slopes different: separants.

3. Heart + · Method [GEO] Show that $A(1;4)$, $B(2;7)$ and $C(4;13)$ are aligned in two ways.

Rep. (i) All verify $y=3x+1$. (ii) $\overrightarrow{AB}\begin{pmatrix}1\\3\end{pmatrix}$, $\overrightarrow{AC}\begin{pmatrix}3\\9\end{pmatrix}$: $\det\left(\overrightarrow{AB},\overrightarrow{AC}\right)=9-9=0$.

4. Back · Flash [PRO] On a scale map $\dfrac{1}{50\,000}$, two cities are 4 cm apart. What is the real distance?

Rep. $4\times 50\,000=200\,000$ cm $=2$ km.